



FUNDAMENTALS OF OPEN SOURCE IT AND CLOUD COM...

WHAT IS LINUX?

The Operating System

Select "Expand" to learn more about the components managed by operating systems such as Linux.

Tasks Performed by an Operating System

We need an operating system to manage components such as...

Processes

Close ^

A process is any actively running instance of a program. On computers with multi-core processors, Linux can manage how to divide the program so that it runs on more than one processor at a time.

Memory Management

Close ^

Linux manages the sharing of internal memory among processes. Allocating and deallocating addresses in RAM is an important aspect of memory management and will happen thousands of times per second.

File Management

Close ^

Linux ensures there is sufficient available memory space to load files from disk to memory and back, enforcing the permissions stored with the data.

Input/Output

Close ^

The Linux kernel handles requests to the hardware like mouse, keyboard, monitor, etc., and ensures the appropriate device driver is loaded for the device.

Interpreting user commands

Close ^

When a user presses a keyboard key, touches a touchscreen or uses a mouse, the operating system will make sure that the machine responds as the user expects.

Resource allocation

Close ^

The term is used to describe the combination of process and memory management to ensure that applications have enough processor time and can be allocated sufficient space in RAM as necessary.

We will expand more on these concepts throughout the course.